Bangladesh Army University of Science and Technology (BAUST), Saidpur

Department of Computer Science and Engineering

Summer 2026 | CSE 2205: Database Management System

SQL Practice Sheet

Scenario

A university stores its departments and students in two tables. Each student belongs to one department through `dept_id`.

Department (dept_id, dept_name, building)

dept_id	dept_name	building
1	CSE	Academic Building-1
2	EEE	Academic Building-2
3	ME	Academic Building-3

Student (student_id, name, email, city, cgpa, dept_id)

student_id	name	email	city	cgpa	dept_id
2205001	Rakibul Islam	rakib@gmail.com	Saidpur	3.75	1
2205002	Nusrat Jahan	nusrat@yahoo.com	Dhaka	3.40	1
2205003	Tanvir Ahmed	tanvir@gmail.com	Rangpur	2.95	2
2205004	Farhana Akter	farhana@baust.edu.bd	Dinajpur	3.85	1
2205005	Mahmudul Hasan	mahmud@gmail.com	Dhaka	2.40	3
2205006	Sadia Rahman	sadia@outlook.com	Rangpur	3.10	2
2205007	Rafiq Uddin	rafiq@baust.edu.bd	Saidpur	3.55	3
2205008	Jannatul Ferdous	jannat@gmail.com	Bogura	3.20	1

Questions

1. Display all the information of all students.
2. Display the name and CGPA of all students sorted by CGPA from highest to lowest.
3. Find the students whose CGPA is greater than 3.50.
4. Find the name and email of students who are from Rangpur and belong to department 2.
5. Find the students whose CGPA is between 3.00 and 3.50.
6. Find the students whose name starts with the letter 'R'.
7. Find the name and email of students who use a Gmail address.
8. Update the CGPA of the student with ID 2205003 to 3.15.
9. The EEE department (dept_id = 2) has moved to 'Academic Building-4'. Update it.
10. Delete all students whose CGPA is less than 2.50.
11. Display each student's ID, name and department name.
12. Display the name, CGPA and building of all CSE students who have a CGPA above 3.50.

Answers

Setup Script (create tables and insert sample data)

```
CREATE TABLE Department (
    dept_id    INT PRIMARY KEY,
    dept_name  VARCHAR(50) NOT NULL,
    building   VARCHAR(50)
);

CREATE TABLE Student (
    student_id INT PRIMARY KEY,
    name       VARCHAR(50) NOT NULL,
    email      VARCHAR(60) UNIQUE,
    city       VARCHAR(30),
    cgpa       DECIMAL(3,2),
    dept_id    INT,
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);

INSERT INTO Department VALUES
(1, 'CSE', 'Academic Building-1'),
(2, 'EEE', 'Academic Building-2'),
(3, 'ME',  'Academic Building-3');

INSERT INTO Student VALUES
(2205001, 'Rakibul Islam',    'rakib@gmail.com',      'Saidpur',  3.75, 1),
(2205002, 'Nusrat Jahan',     'nusrat@yahoo.com',     'Dhaka',    3.40, 1),
(2205003, 'Tanvir Ahmed',     'tanvir@gmail.com',     'Rangpur',  2.95, 2),
(2205004, 'Farhana Akter',    'farhana@baust.edu.bd', 'Dinajpur', 3.85, 1),
(2205005, 'Mahmudul Hasan',   'mahmud@gmail.com',     'Dhaka',    2.40, 3),
(2205006, 'Sadia Rahman',     'sadia@outlook.com',    'Rangpur',  3.10, 2),
(2205007, 'Rafiq Uddin',      'rafiq@baust.edu.bd',   'Saidpur',  3.55, 3),
(2205008, 'Jannatul Ferdous', 'jannat@gmail.com',     'Bogura',   3.20, 1);
```

1. Display all the information of all students.

```
SELECT *
FROM Student;
```

Output:

student_id	name	email	city	cgpa	dept_id
2205001	Rakibul Islam	rakib@gmail.com	Saidpur	3.75	1
2205002	Nusrat Jahan	nusrat@yahoo.com	Dhaka	3.40	1
2205003	Tanvir Ahmed	tanvir@gmail.com	Rangpur	2.95	2
2205004	Farhana Akter	farhana@baust.edu.bd	Dinajpur	3.85	1
2205005	Mahmudul Hasan	mahmud@gmail.com	Dhaka	2.40	3
2205006	Sadia Rahman	sadia@outlook.com	Rangpur	3.10	2
2205007	Rafiq Uddin	rafiq@baust.edu.bd	Saidpur	3.55	3
2205008	Jannatul Ferdous	jannat@gmail.com	Bogura	3.20	1

2. Display the name and CGPA of all students sorted by CGPA from highest to lowest.

```
SELECT name, cgpa
FROM Student
ORDER BY cgpa DESC;
```

Output:

name	cgpa
Farhana Akter	3.85
Rakibul Islam	3.75
Rafiq Uddin	3.55
Nusrat Jahan	3.40
Jannatul Ferdous	3.20
Sadia Rahman	3.10
Tanvir Ahmed	2.95
Mahmudul Hasan	2.40

3. Find the students whose CGPA is greater than 3.50.

```
SELECT *
FROM Student
WHERE cgpa > 3.50;
```

Output:

student_id	name	email	city	cgpa	dept_id
2205001	Rakibul Islam	rakib@gmail.com	Saidpur	3.75	1
2205004	Farhana Akter	farhana@baust.edu.bd	Dinajpur	3.85	1
2205007	Rafiq Uddin	rafiq@baust.edu.bd	Saidpur	3.55	3

4. Find the name and email of students who are from Rangpur and belong to department 2.

```
SELECT name, email
FROM Student
WHERE city = 'Rangpur' AND dept_id = 2;
```

Output:

name	email
Tanvir Ahmed	tanvir@gmail.com
Sadia Rahman	sadia@outlook.com

5. Find the students whose CGPA is between 3.00 and 3.50.

```
SELECT student_id, name, cgpa
FROM Student
WHERE cgpa BETWEEN 3.00 AND 3.50;
```

Output:

student_id	name	cgpa
2205002	Nusrat Jahan	3.40
2205006	Sadia Rahman	3.10
2205008	Jannatul Ferdous	3.20

6. Find the students whose name starts with the letter 'R'.

```
SELECT *
FROM Student
WHERE name LIKE 'R%';
```

Output:

student_id	name	email	city	cgpa	dept_id
2205001	Rakibul Islam	rakib@gmail.com	Saidpur	3.75	1
2205007	Rafiq Uddin	rafiq@baust.edu.bd	Saidpur	3.55	3

7. Find the name and email of students who use a Gmail address.

```
SELECT name, email
FROM Student
WHERE email LIKE '%@gmail.com';
```

Output:

name	email
Rakibul Islam	rakib@gmail.com
Tanvir Ahmed	tanvir@gmail.com
Mahmudul Hasan	mahmud@gmail.com
Jannatul Ferdous	jannat@gmail.com

8. Update the CGPA of the student with ID 2205003 to 3.15.

```
UPDATE Student
SET cgpa = 3.15
WHERE student_id = 2205003;
```

Result: 1 row affected.

9. The EEE department (dept_id = 2) has moved to 'Academic Building-4'. Update it.

```
UPDATE Department
SET building = 'Academic Building-4'
WHERE dept_id = 2;
```

Result: 1 row affected.

10. Delete all students whose CGPA is less than 2.50.

```
DELETE FROM Student
WHERE cgpa < 2.50;
```

Result: 1 row affected.

11. Display each student's ID, name and department name.

```
SELECT s.student_id, s.name, d.dept_name
FROM Student s
INNER JOIN Department d
    ON s.dept_id = d.dept_id;
```

Output:

student_id	name	dept_name
2205001	Rakibul Islam	CSE
2205002	Nusrat Jahan	CSE
2205003	Tanvir Ahmed	EEE
2205004	Farhana Akter	CSE
2205005	Mahmudul Hasan	ME
2205006	Sadia Rahman	EEE
2205007	Rafiq Uddin	ME
2205008	Jannatul Ferdous	CSE

12. Display the name, CGPA and building of all CSE students who have a CGPA above 3.50.

```
SELECT s.name, s.cgpa, d.building
FROM Student s
INNER JOIN Department d
    ON s.dept_id = d.dept_id
WHERE d.dept_name = 'CSE' AND s.cgpa > 3.50;
```

Output:

name	cgpa	building
Rakibul Islam	3.75	Academic Building-1
Farhana Akter	3.85	Academic Building-1